Construction Material Quantity Estimation and digital tracking for Waste Reduction

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Domain: Space, Aerospace, Mobility and Next Gen Infrastructure
Genesis Ideathon 2026

Abstract. Construction material waste is a persistent problem in building projects. Site engineers estimate material quantities from structural design, but the estimate rarely matches the actual need, so extra material is often ordered "just in case". Published studies from Nigeria and Indonesia report actual site waste above the 5% commonly assumed in estimates, with causes such as inaccurate measurements, design changes and over-ordering. A conversation with a mason on a campus construction site also indicated that estimates drift, small waste is rarely recorded, and any tool must be very simple to use. This paper proposes Sustainable AI, an AI-assisted tool that estimates net material quantities from project details, adds a data-driven waste allowance for each material, tracks ordered, used and leftover quantities, and suggests reuse of leftovers. The AI interprets typed or spoken input, standard civil engineering formulas perform the calculations, and the engineer confirms quantities before ordering. Waste factors are initialised from published benchmarks and updated from site records. The concept is at the design stage; a prototype and a wider field study are planned.

Keywords: Material waste · Quantity estimation · Sustainable construction.

1 Problem Statement and Objectives

1.1 Problem Statement

- Site engineers estimate material quantities from structural design, and the estimate rarely matches actual need.
- To avoid work stoppages, extra material is ordered "just in case", for example 100 bags when 70 are needed.
- The surplus becomes cost, storage burden and construction waste.
- Small waste is not measured or recorded, so the problem stays invisible.

1.2 Objectives

- Predict accurate material quantities from project details.

- Add a realistic, data-driven waste allowance for each material.
- Record ordered, used and leftover quantities on site.
- Suggest reuse of leftovers before disposal.
- Learn from every project to improve future estimates.

2 Case Studies

2.1 Case Study 1: Estimator Allowances in Lagos

Oladiran, O.J. (2018), Journal of Sustainable Technology, 9(2), 88–98; 97 building projects, Lagos, Nigeria [4].

Method and scope.

- Site waste measured directly for concrete, blocks, tiles, aluminium profiles and glass.
- Estimators were asked for their waste allowances.
- Regression models were built to predict waste quantity.

Key findings.

- Estimator allowances were 8.36% to 10.06%, not the assumed 5%.
- Glass was the most underestimated material.
- Glass had the highest waste and aluminium profiles the lowest.

Relevance to Sustainable AI: Supports data-driven waste factors and waste prediction models.

2.2 Case Study 2: Waste on Public Projects

Adewuyi, T.O., Idoro, G.I. & Ikpo, I.J. (2014), Civil Engineering Dimension, 16(2), 96–103; South-South Nigeria [1].

Method and scope.

- Data from 30 ongoing public building projects over six months.
- Waste calculated as a percentage per material.
- Paired t-test compared actual waste with allowable waste.

Key findings.

- Measured waste levels ranged from about 10% to 14.5% across materials.
- Actual waste differed significantly from allowances ($p < 0.05$).
- Authors urged adopting measured values in estimates.

Relevance to Sustainable AI: Confirms fixed allowances do not reflect real site waste.

2.3 Case Study 3: Waste Control on Building Sites

Odusami, K.T., Oladiran, O.J. & Ibrahim, S.A. (2012), Evaluation of materials wastage and control, Nigeria [3].

Method and scope.

- Questionnaire survey of construction professionals on selected building sites.
- Descriptive statistics used to analyse wastage and control strategies.

Key findings.

- Estimators' allowances were exceeded for most materials.
- Block work had the highest level of waste generation.
- Site supervision and control was the most important strategy.

Relevance to Sustainable AI: Shows monitoring on site is a key lever, which our tracking module supports.

2.4 Case Study 4: Four-Storey Office Shop Building

Asmi, A., Munandar, T.V. & Djamaris, A. (2024), International Journal of Sustainable Construction Engineering and Technology, 15(4), 38–52 [2].

Method and scope.

- On-site questionnaires analysed with SPSS.
- Compared budget estimates, daily reports and purchase records.
- Applied the waste hierarchy concept.

Key findings.

- Causes: inaccurate on-site measurements, complex design changes, design errors.
- Most wasted: concrete, bricks and reinforcing steel.
- Recommended accurate estimates, tracking and reuse of leftovers.

Relevance to Sustainable AI: Directly matches our three modules: estimate, track and reuse.

2.5 Case Study 5: Change Orders in Road Projects

Sulistio, H. & Waty, M. (2024), Planning Malaysia, 22(2), 533–549; 700 respondents [7].

Method and scope.

- Interviews and a questionnaire built with the Delphi method.
- Respondents: owners, consultants and contractors.
- Factor analysis and PLS-SEM modelling.

Key findings.

- Change orders explained 73.3% of the modelled waste impact.
- Top indicators: slow drawing revisions, on-site damage, inaccurate measurements, wrong storage.
- Ordering errors causing excess or shortage were identified.

Relevance to Sustainable AI: Design changes must be easy to update so quantities stay current.

2.6 Case Study 6: Lean Construction on a Residential Building

Sapitri, Oktaviani, A. & Jusoh, M.N.H. (2026), Evergreen, 13(02), 851–862 [6]. Waste level = waste volume / installed material × 100.

Table 1. Waste level by material (%).

Material	D19	D16	D13	D10	K-300	Ø12	Ø10	Ø8
Waste level (%)	1	11	3	5	3	5	1	0

Key findings.

- D16 rebar waste reached 11%, above the usual 5% limit.
- Overuse of material beyond need was the top cause.
- Contractors supplied extra material than planned.
- Suggested staged delivery and BIM.

2.7 Case Study Synthesis

Table 2 summarises the six studies.

Table 2. Case study synthesis.

Study	Setting	Main finding
Oladiran (2018)	97 projects, Lagos	Allowances 8.36–10.06% vs 5% assumed
Adewuyi et al. (2014)	30 public projects	Waste about 10–14.5%, above allowable
Odusami et al. (2012)	Building sites, Nigeria	Allowances exceeded; supervision is key
Asmi et al. (2024)	Office shop building	Measurement errors and design changes
Sulistio & Waty (2024)	Road projects	Change orders drive waste
Sapitri et al. (2026)	Residential building	D16 rebar 11%; overuse is top cause

Common pattern: fixed allowances underestimate real waste, and measurement errors, design changes and over-ordering drive it. Few studies offer a simple tool for site engineers. A further study analysed waste level and cost for steel reinforcement work [5].

3 Field Conversation with a Mason

Campus construction site near Kashyapa Bhavanam, Amrita Vishwa Vidyapeetham, Coimbatore. This is a single interview and is indicative, not statistically representative.

Me: Anna, how do you get cement, sand and bricks for the work?
Mason: The engineer tells us how much to bring. We bring it and keep it near the work.

Me: Does that amount always come out right?
Mason: No, sometimes it is less and sometimes more. The work may change, or we may need a little extra mortar. If it runs out, we have to wait for more.

Me: What do you do with the materials left over?
Mason: Good bricks, we keep and use again. If sand is clean, we use it. Broken bricks and dried mortar are put aside. Sometimes the leftover concrete is also thrown away after it hardens.

Me: Do the materials get spoiled?
Mason: Yes. If rainwater gets into the cement, it becomes hard and we cannot use it properly. Sand can also get mixed with mud if it is left on the ground.

Me: Do you write down how much material is wasted each day?
Mason: No, not every little bit. We see what has come and what we have used. Small waste is usually just put aside or thrown away. The engineer may keep the main account.

Me: What happens if there is too much or too little material?
Mason: If it is too little, we stop the work and wait. If there is too much, it takes up space. Cement can get spoiled if it is kept for too long or gets wet.

Me: Would a mobile app for estimating materials and tracking leftovers be useful?
Mason: It could be useful, especially for the engineer or supervisor. It should be simple to use, and updating the quantity used or left over shouldn't take too much time. If it's complicated, workers may not use it regularly.

3.1 Field Findings and Design Response

4 Mechanism: How Sustainable AI Works

1. **Input:** The engineer types, speaks or uploads project details such as dimensions, grade and element type.

Table 3. Field findings and design response.

Field finding	Design response
Estimates drift as work changes	Quantities are recalculated when dimensions or design change.
Small waste is not recorded	One-tap logging of used, leftover and wasted quantities.
Rain and mud spoil material	Storage and staged-delivery reminders in the order plan.
Workers need simplicity	Minimal screens; engineer or supervisor updates data.

2. **AI Interpretation:** The AI extracts the values, checks for missing details and asks clarifying questions.
3. **Formula Engine:** Standard civil calculations convert dimensions into net material quantities.
4. **Waste Prediction:** A material-specific waste factor, learned from site data, is added to the net quantity.
5. **Order Plan and Feedback:** Staged order quantities are issued; actual usage is logged to improve the next estimate.

4.1 Calculation and Learning Logic

Net quantity. Computed from dimensions, for example

$$\text{Volume} = \text{Length} \times \text{Breadth} \times \text{Depth}. \tag{1}$$

Order quantity.

$$Q_{order} = Q_{net}\,(1 + w), \tag{2}$$

where w is the waste factor.

Waste level (%). As used by Sapitri et al. (2026) [6],

$$W = \frac{\text{Purchased} - \text{Installed}}{\text{Installed}} \times 100. \tag{3}$$

Surplus (%). Logged from site records,

$$S = \frac{\text{Ordered} - \text{Used}}{\text{Ordered}} \times 100. \tag{4}$$

The AI interprets input; formulas do the math; the waste factor starts from published benchmarks and is updated from each site's logs.

5 Expected Impact and Roadmap

5.1 Expected Impact

- Lower material cost through accurate ordering.
- Fewer shortage delays and less idle time.
- Less storage burden and disposal.
- Illustrative example: Rs 10 lakh of material with 5% less waste saves Rs 50,000 per project. Real figures will follow our field study.

5.2 Roadmap

- Phase 1: Wall, slab and column quantity calculator (MVP).
- Phase 2: Usage logging and leftover tracking.
- Phase 3: Interviews at more sites to calibrate waste factors.
- Phase 4: Drawing and BIM input, supplier ordering, leftover exchange between sites.

6 Conclusion

Order only what you need. Sustainable AI combines evidence from six studies and a field conversation to estimate accurately, track usage and reuse leftovers, reducing cost and construction waste.

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